

LEONARDO SANTOS

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[Homepage](#) ◊ [Google Scholar](#) ◊ [Github](#) ◊ [LinkedIn](#)

EDUCATION

KTH Royal Institute of Technology

August 2025-Present, Sweden

Ph.D Student in the Division of Robotics Perception and Learning (RPL)

Supervisor: Prof. Iolanda Leite, Co-supervisor: Prof. Jana Tumova.

UFMG Federal University of Minas Gerais

July 2025, Brazil

B.Sc. in Electrical Engineering, Emphasis in Robotics

GPA: 86/100

SKILLS

Programming Python, C++, MATLAB

Technologies ROS, RViz, Git, $\text{\LaTeX}$

Languages English (fluent) and Portuguese (native)

RESEARCH EXPERIENCE

Graduate Research Assistant

August 2025 – Present

KTH | [Prof. Iolanda Leite](#) | [RPL](#)

Stockholm, Sweden

- Doing research on interpretable reward learning in RLHF.

Robotics Institute Summer Scholar (RISS)

June 2024 – August 2024

Carnegie Mellon University | [Prof. Andrea Bajcsy](#) | [Intent Lab](#)

Pittsburgh, USA

- Developed a framework to update safety specifications online from human language feedback using a Vision Language Model.
- Applied the safety specifications in a downstream reachability controller for safe navigation and demonstrated the system in both hardware and simulation experiments. Co-first author [paper](#) under review to ICRA 2025 [1].
- More information and demos on the [project website](#).
- **Technical skills:** Python, ROS.

Research Assistant

September 2023 – January 2024

KTH | [Prof. Jana Tumova](#) | [RPL](#)

Stockholm, Sweden

- Studied risk-aware trajectory optimization for safe navigation among humans.
- Implemented an MPPI-based trajectory optimizer to minimize a CVaR risk measure in belief space for trajectory planning under uncertainty.
- **Technical skills:** Python, PyTorch.

Robotics Institute Summer Scholar (RISS)

June 2023 – August 2023

Carnegie Mellon University | [Prof. Sebastian Scherer](#) | [AirLab](#)

Pittsburgh, USA

- Developed a framework based on Random Finite Sets that mathematically unifies search and track objectives for informative path planning.
- Contributed to both theoretical and implementation developments.
- Co-first author [paper](#) published at 2025 AIAA SciTech Forum [2].
- **Technical skills:** C++, MATLAB, ROS.

Research Assistant

UFMG | [Prof. Douglas Macharet](#) | [VeRLab](#)

February 2021 – May 2023

Belo Horizonte, Brazil

- Developed a planning pipeline for communication-constrained multi-robot systems. The pipeline is based on virtual structures and guarantees that communication between robots is preserved by construction. Demonstrated in simulation and in a simple hardware experiment. First author [paper](#) under review at IEEE Latin America Transactions [3].
- Ran and evaluated experiments in a project that uses a heterogeneous multi-robot system for warehouse management. Second author [paper](#) published at LARS 2022 [4].
- **Technical skills:** Python, ROS.

PUBLICATIONS

- [1] L. Santos*, Z. Li*, L. Peters, S. Bansal[†], and A. Bajcsy[†], “Updating robot safety representations online from natural language feedback,” *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- [2] L. Santos*, B. Moon*, S. Scherer, and H. Van Nguyen, “Unisat: Unified-objective belief model and planner to search for and track multiple objects,” in *AIAA SCITECH 2025 Forum*, 2025, p. 2114.
- [3] L. Santos, C. C. G. Ribeiro, and D. G. Macharet, “Communication backbone reconfiguration with connectivity maintenance,” *arXiv preprint arXiv:2409.16851*, 2024.
- [4] C. C. G. Ribeiro, L. H. M. C. dos Santos, and D. G. Macharet, “Collaborative ugv/uav path planning for inventory management in warehouses,” in *2022 Latin American Robotics Symposium (LARS)*, 2022, pp. 121–126.

* Indicates equal contribution, † indicates equal advising.

AWARDS, GRANTS AND SCHOLARSHIPS

IEEE RAS Travel Grant to attend ICRA 2025	<i>Spring 2025</i>
Google Undergraduate Research Scholarship , 1 of 3 scholars at CMU	<i>Summer 2024</i>
Minas Mundi Study Abroad , selected UFMG student	<i>Fall 2023</i>
Mobility21 Undergraduate Research Scholarship , 1 of 3 scholars at CMU	<i>Summer 2023</i>
National Finalist , top 3 Overall, 2nd SARC-BARINet Aerospace Competition	<i>Fall 2022</i>
Top 15 Papers Award out of 80 submissions, LARS 2022	<i>Fall 2022</i>
Minas Gerais Research Support Foundation Scholarship	Fall 2022, Spring 2023
Brazilian National Council for R&D Scholarship	Fall 2021, Spring 2022

SERVICE AND OUTREACH

Reviewer July 2025 – Present

- NeurIPS 2025 (Position paper track); ACM/IEEE HRI 2026.

Member of Admissions Committee January 2025

[Carnegie Mellon University's Robotics Institute Summer Scholars](#)

- Reviewed applications for the Robotics Institute Summer Scholars (RISS) program at Carnegie Mellon University (CMU).
- RISS is a funded summer research program that brings national and international students to do research at the CMU Robotics Institute.

Co-Lead Organizer June 2023 – August 2024

[RoboLaunch](#) Pittsburgh, USA

- Helped organize a robotics seminar series aimed at introducing undergraduate students to robotics through a series of accessible talks from speakers from academia and industry.
- Led logistics involving speaker invitations, advertisement, and general team coordination.

- During the time I was involved, we hosted speakers from CMU, MIT, the University of Washington, the University of Adelaide, ETH Zurich, among others.

Co-Founder and Webmaster
UFMG IEEE Student Branch

September 2024 – July 2025
Belo Horizonte, Brazil

- Organized events to promote and advertise research at UFMG, including seminars on microelectronics, AI, robotics, and biomedical engineering; research partnerships between Brazilian and French universities.
- Was responsible for everything related to branding, external communications, and social media presence.

SOCIETY MEMBERSHIPS

IEEE RAS Student Member
IEEE Student Member

2025 – Present
2024 – Present